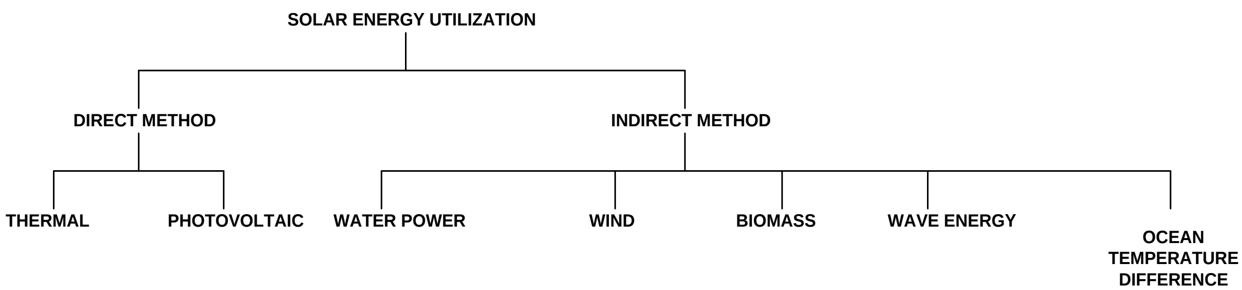
SOLAR INSTALLATION SYSTEMS FUNDAMENTALS

INTRODUCTION

Solar energy is the most abundant energy resource on earth. The solar energy that hits the earth's surface in one hour is about the same as the amount consumed by all human activities in a year. Direct conversion of sunlight into electricity in photovoltaic (PV) cells is one of the three main solar active technologies, the two others being concentrating solar power (CSP) and solar thermal collectors for heating and cooling (SHC). Today, PV provides 0.1% of the total global electricity generation. However, PV is expanding very rapidly due to effective supporting policies and recent dramatic cost reductions. PV is a commercially available and reliable technology with a significant potential for long term growth in nearly all world regions.

Kenya's geographical location astrides the equator gives a unique opportunity for a vibrant solar energy market. The country receives good solar insolation all year round coupled with moderate to high temperatures estimated at 4-6kWh/m^2/day.



Definition of terms

Radiation

This is the movement of energy through a vacuum

Direct Radiation

This is the solar radiation received at the earth's surface without change of direction from the sun. It comes in straight beam and can be focused by a lens or mirror.

Diffuse or Indirect radiation

This is the radiation received at the earth's surface from all parts of the sky's atmosphere after being subjected to scattering in the atmosphere. Clouds and dust absorb and scatter the radiation.

Global Radiation

This is the sum of direct and diffuse radiations.

Solar Irradiance

This is the solar power received per unit area. It is measured in watts per square metre (W/m^2). Maximum irradiance occurs when the solar incident angle is 90^0

Radiance existence/Radiance emittance

This is bhthe solar power emerging from a unit surface area. It is measured in watt per square metre (W/m^2)

Insolation

This is the solar energy received on a specified area over a specified period. It is measured in kilowatt-hours per square metre per day (kWh/m^2 per day)

Peak sun hours

This is the equivalent number of hours each day when the solar irradiance averages $1000W/m^2$

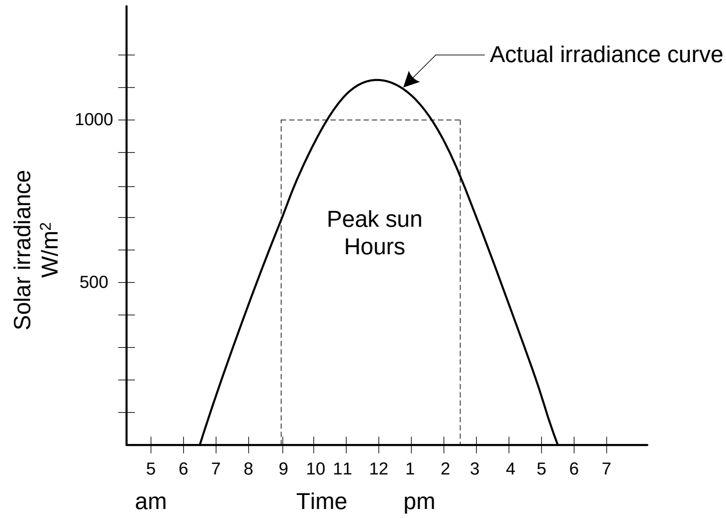


Figure 6.01 Solar irradiance received over time on a flat surface.

Solar constant

This is the quantity of solar power (W/m^2) at normal incidence outside the atmosphere (extraterrestrial) at the mean sun-earth distance. Its mean value is 1377.7W/m^2. Only about 1000W/m^2 reaches the earth's surface due to scattering of waves by the ozone layer, cloud, moisture and dust.

CONVERSION OF SOLAR ENERGY

Solar energy can be converted to other forms of energy

Solar energy to chemical energy

Plants convert solar energy into chemical energy through a process called photosynthesis.

Sucrose is made for plants growth.

Solar energy to heat energy

Radiant energy is directed using solar reflectors or concentrators and collected as heat by trapping and absorbing it on specially treated surfaces.

Solar energy to biomass

Plant matter created by the process of photosynthesis is called biomass. The chemical equation is shown below.

$$H_2O + CO_2 \rightarrow CH_1O + O_2$$

Biomass includes all plant life, trees, agricultural plants bushes, algae and grass and their residues after processing. It includes animal and human waste.

Ways of obtaining energy from biomass

a) Direct method

Example is burning of wood

b) Indirect method

Biomass can be used indirectly by converting it into a convenient useable fuel in solid, liquid or gaseous form. This conversion can be in two forms:

i) Thermochemical conversion
This includes destructive distillation, pyrolysis and gasification

ii) Biological conversion
Includes the process of fermentation (biomass through anaerobic fermentation)

METHODS OF SOLAR ENERGY HARVESTING.

(a) Parabolic Reflectors/Cylindrical Parabolic Concentrating Collectors

Solar concentrators and reflectors redirect solar radiation incident over a large area by focusing it to a small area. It is then trapped and harnessed at high temperatures.

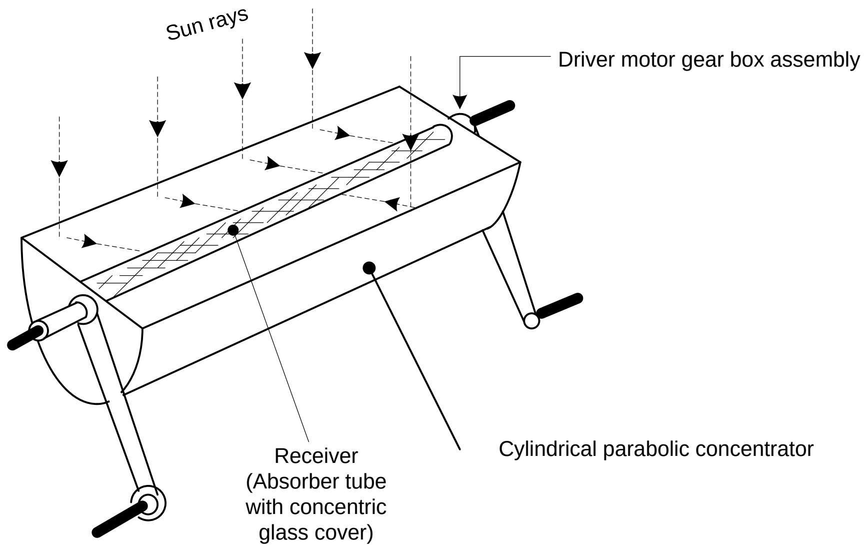


Figure 6.02 Parabolic trough

The collector consists of a conductor and a receiver. The concentrator consists of a mirror reflector having the shape of a cylindrical parabola.

OPERATION.

The concentrator focusses the sunlight onto its axis where it is absorbed on the surface of the absorber tube and transferred to the fluid flowing through it. A concentric glass cover around the absorber tube helps in reducing the convective and radiative losses to the surrounding. The concentrator has to be rotated in order that the suns' rays are always focused onto the absorber tube. This movement is called tracking. Rotation about a single axis is generally required. Fluid temperatures up to 400^0C can be achieved in a cylindrical parabolic focusing collector systems.

(b) Dish-Type Concentrator/Paraboloid Concentrating Collector.

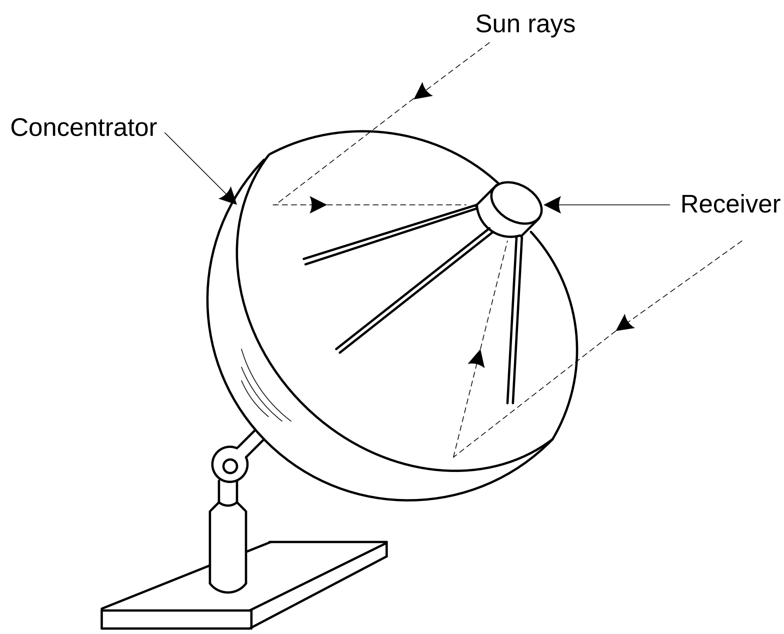


Figure. 6.03 Dish-type concentrator

Uses a single-point focus

Has two axis tracking to ensure that the sun is always in line with the focus and vertex of the paraboloid.

(c) Flat- Plate Collector.

The greenhouse Effect

This allows the accumulation of heat from solar energy. Radiant energy can pass through glass (or clear plastic) surfaces but infrared (heat) energy cannot.

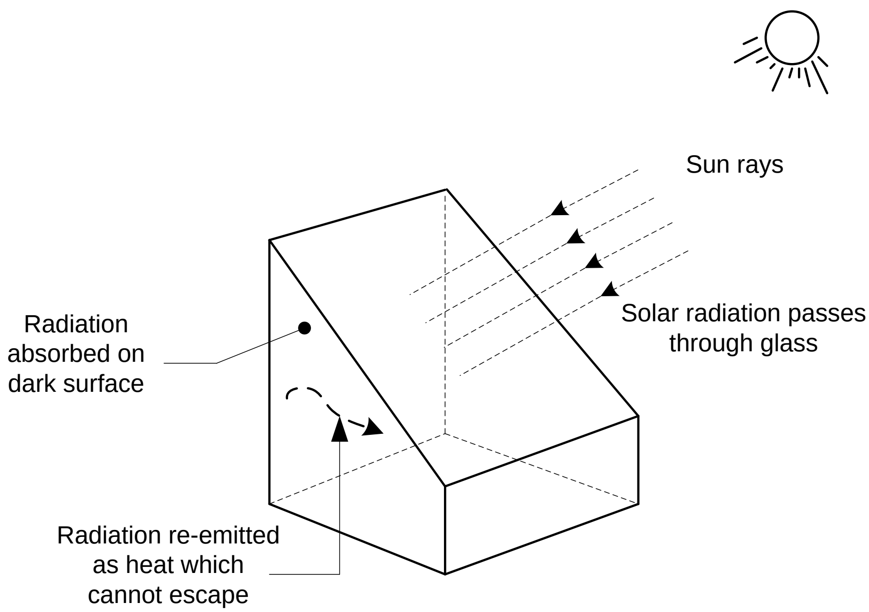


Figure 6.04 Greenhouse effect

Solar radiation passes through the glass windows of a seal box and is absorbed by the surface behind the glass and re-radiated as heat which cannot pass through the glass. This heat causes the temperature to rise inside the box. Flat plate collectors take advantage of both greenhouse effect and absorption principles to trap solar energy.

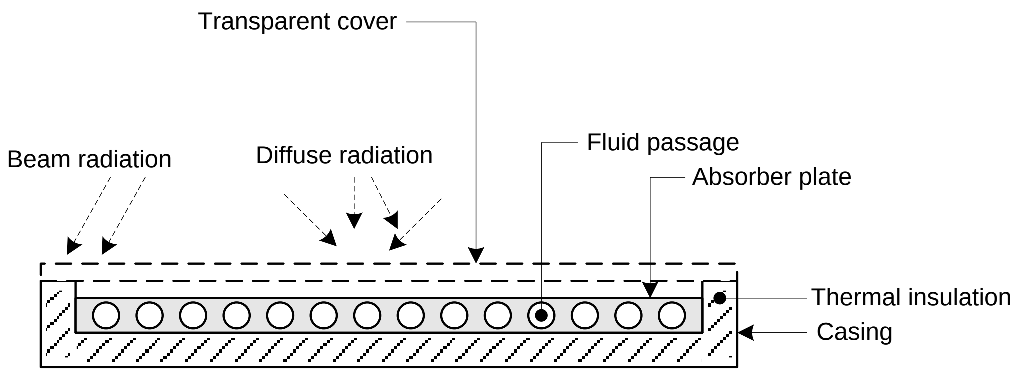


Figure 6.05. Cross sectional view of flat-plate collector.

Flat – plate collectors are designed to heat water to high temperatures and could reach approximately 100^{0}C.

Solar radiation passes through the transparent cover and is absorbed by the absorber plate and also transferred to the liquid flowing through the tubes. The energy transferred to the liquid is the useful gain. The transparent cover helps reduce the heat loss by re-radiation and convection. Thermal insulation reduces heat loss by conduction. Water is the common liquid used. Temperatures of 40^0 C to 100^0 C can be achieved.

A typical flat-plate collector size is 1 m x 2 m.

A flat plate collector has the following parts;

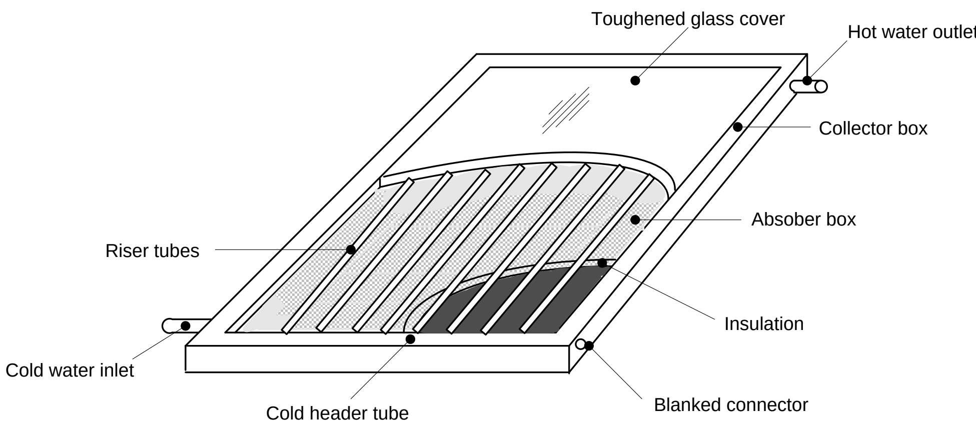


Figure 6.06. Parts of flat-plate collector.

i) *Collector box*

A box or frame that holds all the components together. The material is chosen for its environmental durability as well as its structural characteristics

ii) Transparent Collector cover-

A transparent glass covers over the collector box that allows the sun's rays to pass through to the absorber plate. In general, the sun's heat energy passes quite easily through the collector cover as short waves. Heat energy that re- radiates from the collector absorber is in long waves. Glass prevents long waves from passing back through the cover. This is effectively an example of the greenhouse effect and in this instance; it helps the collector box to stay warm

The glass cover also serves as a transparent insulation layer and is typically a single layer of glass. This reduces air motion across the absorber plate, thereby reducing heat loss through conversion.

iii) Insulation

Material between the absorber plate and the surfaces it touches the blocks heat loss by conduction thereby reducing the heat loss from the collector box. The insulation must be able to withstand extremely high temperatures, so insulations like Styrofoam are not suitable.

Materials that are good insulators and are suitable for this purpose are rock wool or fiber-glass. Typical insulation thickness is around 50mm.

iv) Absorber plate

A flat, usually metal surface inside the collector box that, because of its physical properties, can absorb and transfer high levels of solar energy. The absorber material may be painted black with a semi- selective coating or it may be electroplated or chemically coated with a spectrally selective material (selective surface) that increases its solar absorption capacity by preventing heat from re-radiating from the absorber plate.

v) Rise- tubes

High conductive metal tubes (usually copper) across the absorber through which fluid flows, transferring heat from the absorber to the fluid. The heat transfer fluid tubes removes heat from the absorber plate. The collector may also incorporate headers, which are the fluid inlet and outlet tubes. Header tubes can also referred to as the manifold.

Advantages of flat-plate collector

1. Simple design
2. No moving parts
3. Little maintenance

(d) Photovoltaic/Solar Cells

PV or solar cells operate by converting light energy into electric energy. This action occurs in all semiconductors which are constructed to absorb energy. Photovoltaic (PV) technology converts solar energy into electricity without any moving parts, consuming no conventional fossil fuels, creating no pollution, and lasting for decades with very little maintenance. Based on semiconductor technology, solar cells operate on the principle that electricity flows between two semiconductors when they are put into contact with each other and exposed to light (photons).

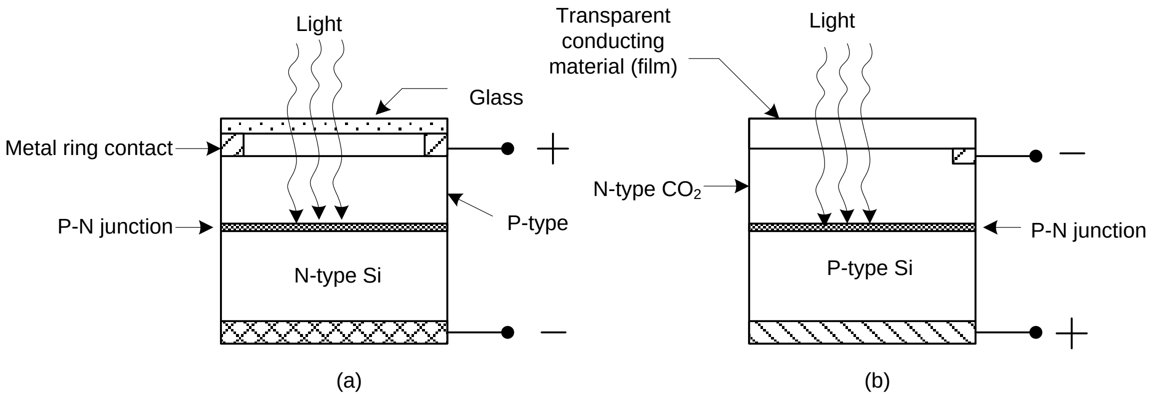


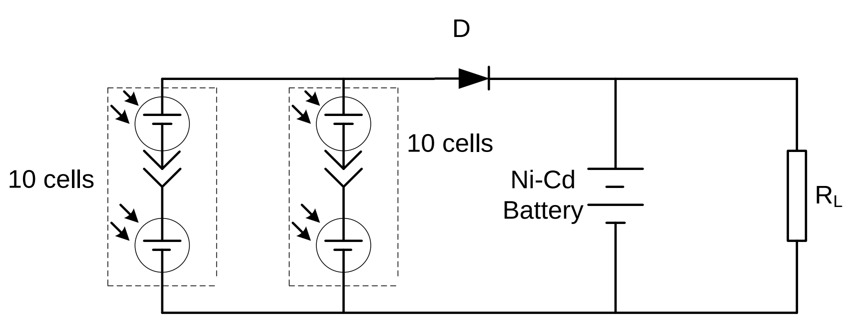
Figure. 6.07 Construction of solar cells

In diagram (a)Light passes through the thin p-type material (0.5mm). This results in the emission of electrons and a buildup of a strong electric field around the pn junction. The emitted electrons are conducted via the continuous conducting surface connected to the N-type material through an external load.

In diagram (b)

In this case the emitted electrons are conducted via the copper negative terminal. The maximum efficiency of the cell in converting sunlight to electricity is about 15%. The terminal voltage is dependent on the amount of light falling on the cell. It can be as high as 0.6V. Large number of cells can be arranged in an array to obtain a higher voltage and current.

Example



Two groups of 10 series cells connected in parallel to each other. If each cell has a voltage of $0.5V$ at $150mA$. Calculate the overall voltage and current in the solar cell.

Solution.

$Voltage\ (series) = 0.5 \times 10 = 5V$

Maximum current from the two parallel branches$= 150 \times 2 = 300mA$.

Therefore the overall rating will be $5V, 300mA$

APPLICATION OF SOLAR ENERGY.

i. Crop drying
ii. Cooking
iii. Water heating
iv. Electricity
v. Space heating
vi. Green houses
vii. Space cooling and refrigeration
viii. Distillation

Water Heating

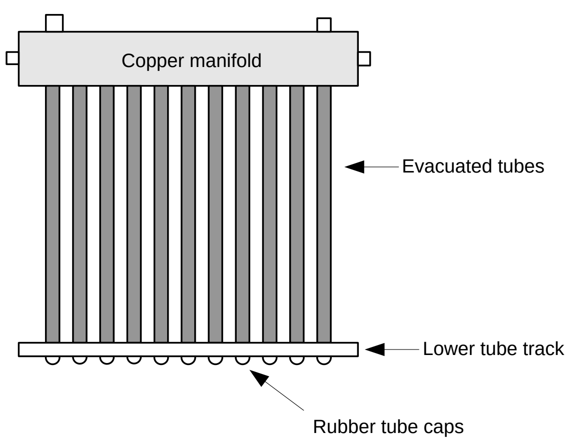
This is an economical water heating application because it uses free energy from the sun.

Main components are;

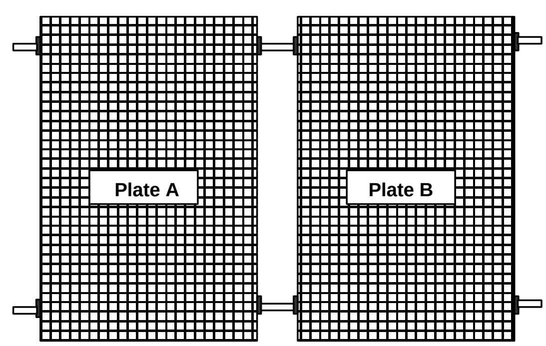
i. Collector

There are two types of collectors namely, Flat plate and Evacuated tubes collector.

The illustration is as shown in figure 6.08



Evacuated tubes connection



Flat plate tubes connection

Figure 6.08 Collector types

ii. Storage tank

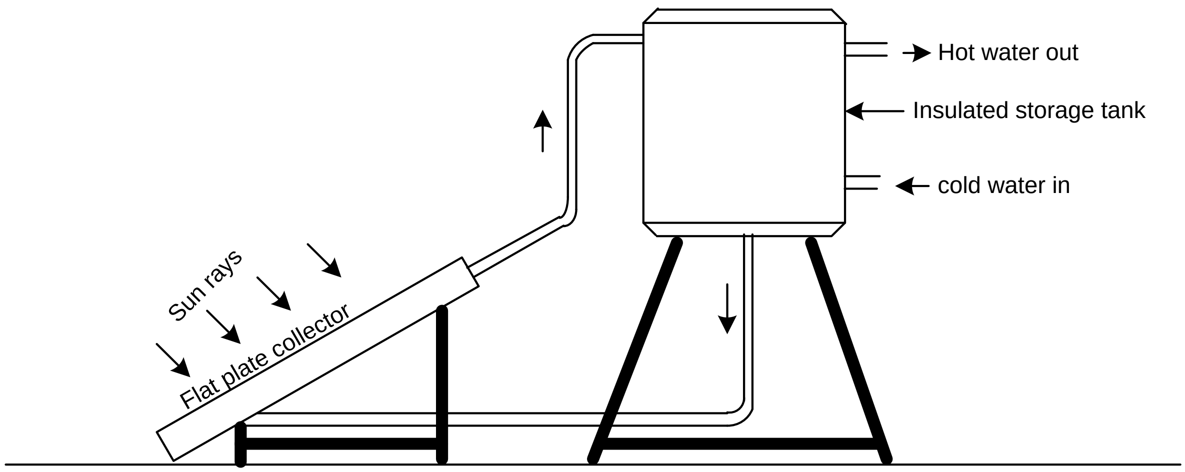


Figure.6.09 Small capacity natural circulation (Thermosiphon) water heating system

Water in the flat-plate collector is heated and automatically flows to the top of the storage tank. It is replaced by water from the bottom of the tank. The hot water is withdrawn for use from the top of the tank and is replaced by the cold water from the bottom

SOLAR WATER HEATING SYSTEMS

There are two main types of solar water heating systems

(i) *Passive solar water heating systems and*

(ii) *Active solar (Split) water heating systems*

Passive Solar Water Heating System

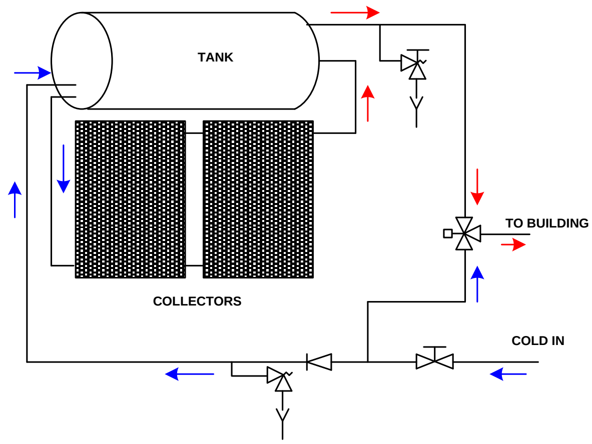


Figure 6.10 Passive solar water heating system

This is the most common SWH system for residential use in Kenya. The cold water from the main water connection first enters through the isolation valve and goes to the non- return valve. The connection between these two is tapped and fed to the thermostatic mixing valve (or tempering, valve) to reduce the temperature of heated water from the tank before the final use.

The cold water inlet after the non-return valve has an optional expansion valve which works as a secondary line of protection in case the temperature pressure relief (TPR) valve fails to operate due to scaling problems. The cold water then enters from the bottom of the tank to ensure that it doesn't mix with the existing hot water that might be present in the tank. Water then flows through the collector loop to the collectors.

The cold water leaving the tank towards the collectors is referred to as "cold flow ". It enters the bottom of the collectors along the lower manifold and rises up through the collector as it is heated by the sun's energy. This is called the Thermosiphon effect; it causes the heated water to rise. The piping in the collector loop from the top manifold to the middle of the tank is called the "hot return ".

Hot water is drawn from the tank for use in the building through an outlet located at the top of the tank. At this point a TPR valve is installed to relieve excessive temperature and pressure in the tank.

The hot water finally enters the thermostatic mixing valve (or tempering valve) where it gets mixed with the cold water to give tempered water at around 500C

Active Solar Water Heating System

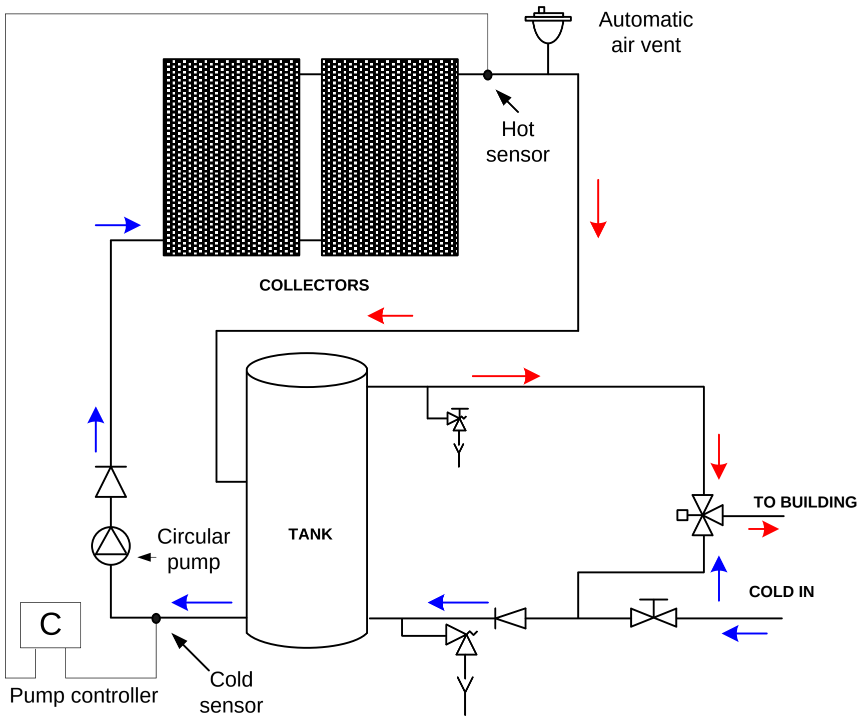


Figure. 6.11 Active Solar Water Heating System

The cold water from the mains water connection enters through the isolation valve and goes through the non-return valve. Tapping is done in between the two valves and fed to the thermostatic valve which mixes the heated water and the cold water to reduce temperature entering the house.

After the cold water non-return there is expansion valve which protects the system incase pressure relief valve fails to operate. The cold water enters from the bottom of the tank to avoid mixing with the already hot water in the tank.

Water is the pumped through the collector loop, enters the bottom of the collectors along lower part and pumped through the collector where it is heated by the suns energy.

Space Heating

Space heating is used is used to keep rooms hot by hot air. The water is in the flat plate collector. This water is pumped from the storage tank. The heated water is pumped to the heat exchanger where it heats incoming cold air.

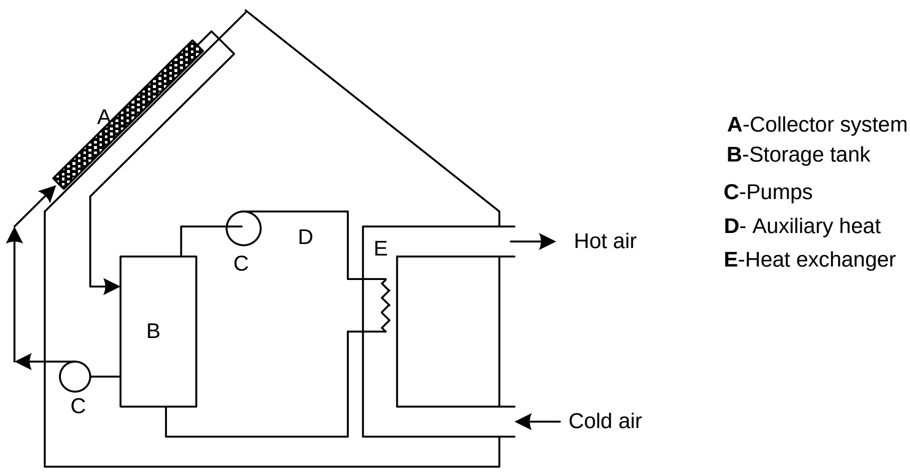


Figure.6.12 Space heating

The hot air flows into the room drawing in cool air into the heat exchanger.

The water in the exchanger losses heat and flows to the storage tank for the cycle to be repeated.

Distillation

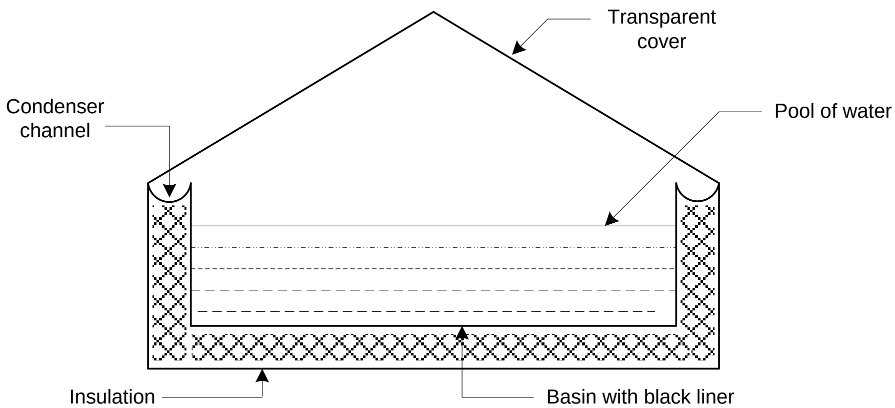


Figure 6.13 Distillation

The distiller consists of a shallow airtight basin lined with a black impervious material which contains salty water. A sloping transparent cover is provided at the top.

Operation

Solar radiation is transmitted through the cover and is absorbed by the black lining. It also heats up the water by about 10^0 to 20^0C and causes it to evaporate. The resulting vapour rises, condenses as pure water on the underside of the cover and flows into condensation collector channels on the sides.

Drying

This removes moisture and helps in preservation of the produce. Traditionally drying is done in open ground. Open ground drying has the following disadvantages

- Slow process
- Insect and dust get mixed with the produce

The use of driers helps to eliminate these disadvantages. Drying is done faster in a controlled manner. A better quality product is obtained.

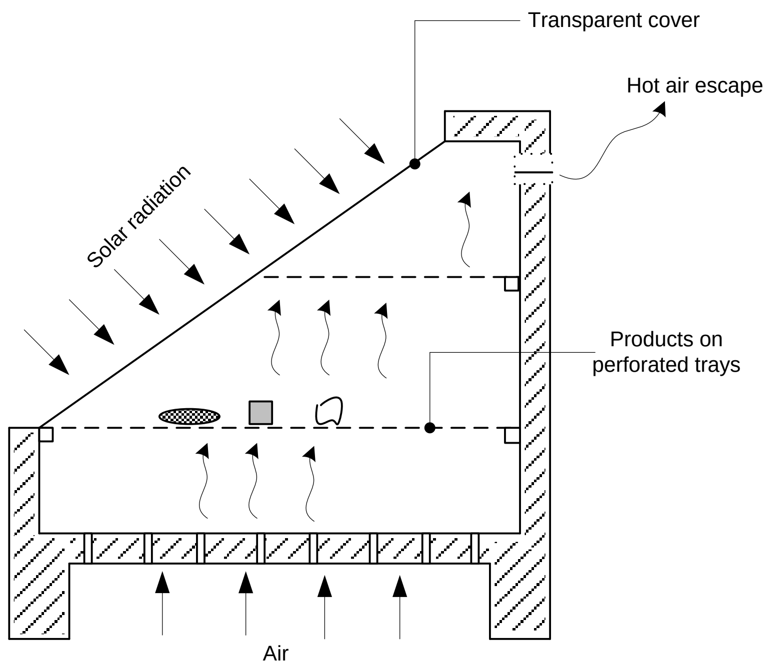


Figure 6.14 Drier

The material to be dried is placed on perforated trays. Solar radiation entering the enclosure is absorbed in the product itself and the surrounding internal surfaces of the enclosure. As a result, moisture is removed from the product and the air inside is heated. Suitable openings at the bottom and top ensure a natural circulation.

Temperatures ranging from 50^0C to 80^0C are usually attained and the drying time ranges from 2 to 4 days.

Cooking

a) Box type cooker

This consists of a rectangular enclosure insulated on the bottom and sides and having one or two glass covers on the top. Solar radiation enters through the top and heats up the enclosure in which the food to be cooked is placed in shallow vessels. Temperatures around 100^0C can be obtained in these cookers on sunny days

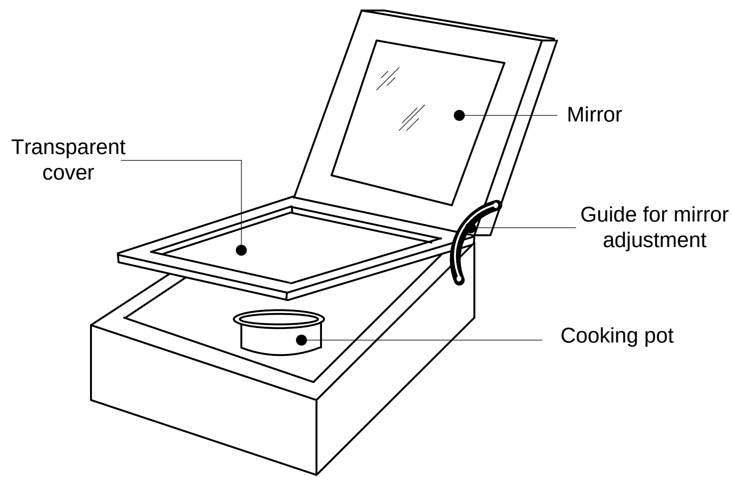


Figure 6.15 Box cooker

Green House

Infrared radiations are used in green houses to grow crops. Shorter wavelength infraredor rays and visible rays from the sun pass through the glass or transparent canvas or plastic panes of a greenhouse raising the temperatures within. They are absorbed by the soil and plants. The soil and plants in turn emit longer wavelength infrared rays which cannot penetrate the glass/plastic. The inside of the glass/plastic house remains warm. (Greenhouse effect)

Electricity

This is achieved by using photovoltaic cells (solar cells). The cells convert the solar radiation (light energy) to electrical energy. The energy is stored as chemical energy in a battery.